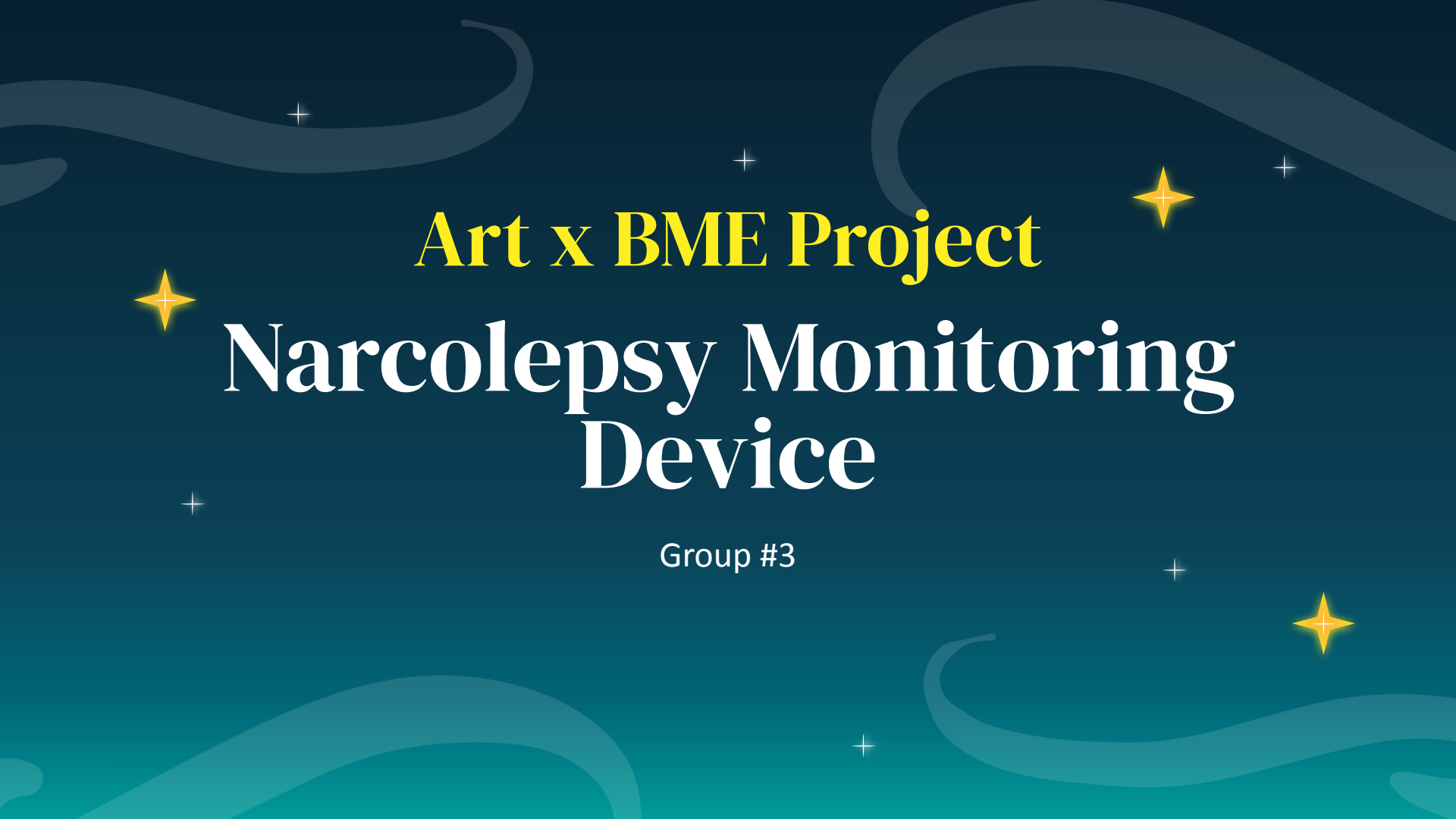




Art x BME Project

Narcolepsy Monitoring Device

Group #3

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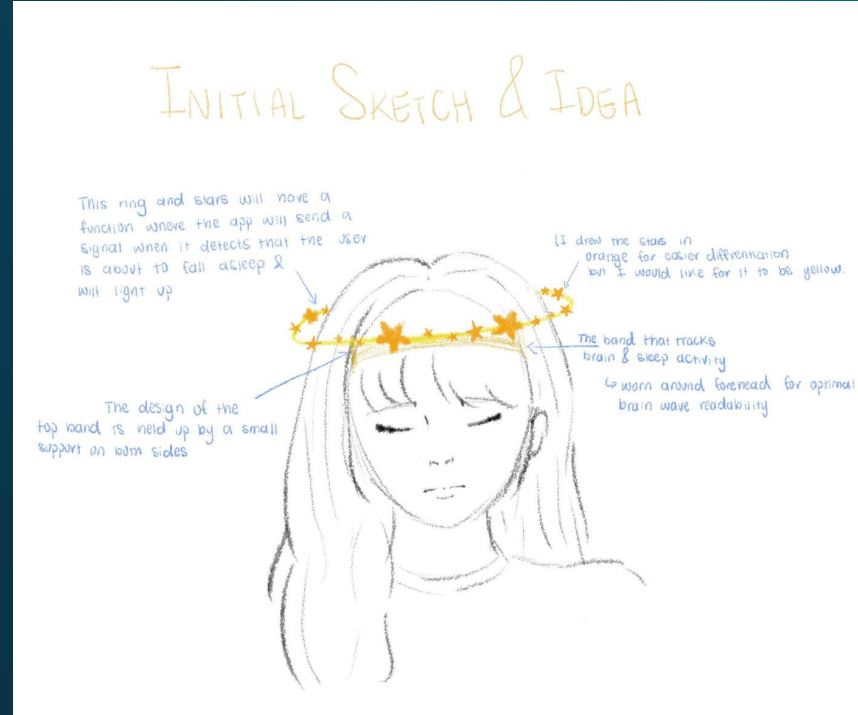
The story:

We wanted to create a narcolepsy monitoring headband that is suitable for children in order to track sleep and brain activity throughout the day.

As there are no current wearable devices for helping with narcolepsy, this device would be able to help keep track of sleep and brain activity using electroencephalogram (EEG) sensors to measure brain waves.

We wanted it to contain a feature that allows for others to know when the user is about to fall asleep, as well as an app that allows for the data regarding what the device collects to help the child's guardian understand the patterns of the child's narcolepsy.

Original Sketch:

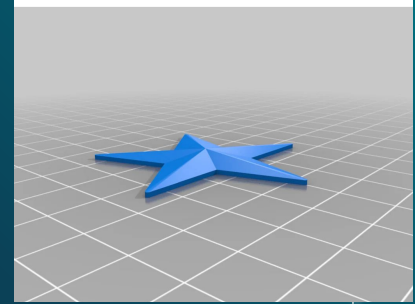
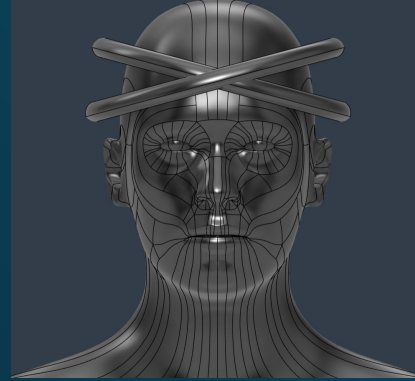


Engineering Aspects of This Project

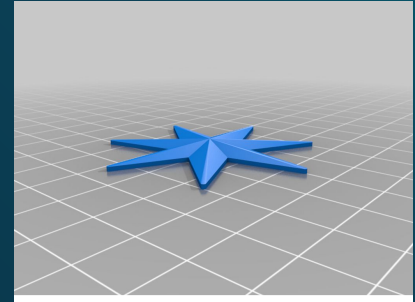
3D design requirements and resources:

- The design of our headband must fit our head
 - 20cm circumference model
 - Non Perfect circle shape
- Thingiverse file for stars [1]

3D Models in Fusion:



Final Product:

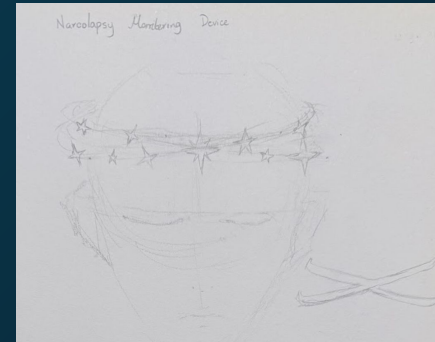
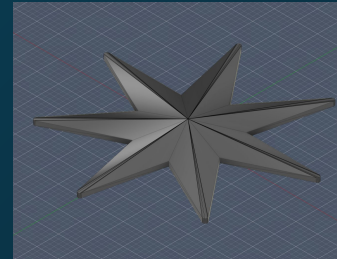


Challenges and Solutions:

Issues / Complications:

- Scheduling conflicts between BME and Art students
 - Frequent communication + using locker
- Printed headband being too large for model
 - Different angle wearing
- Design changes over time

Progress Images



Overall Project Feedback

Feedback:

Our project shifted more towards the art students' ideas and vision

- Double banded
- Golden stars
- Gold + dark green decals

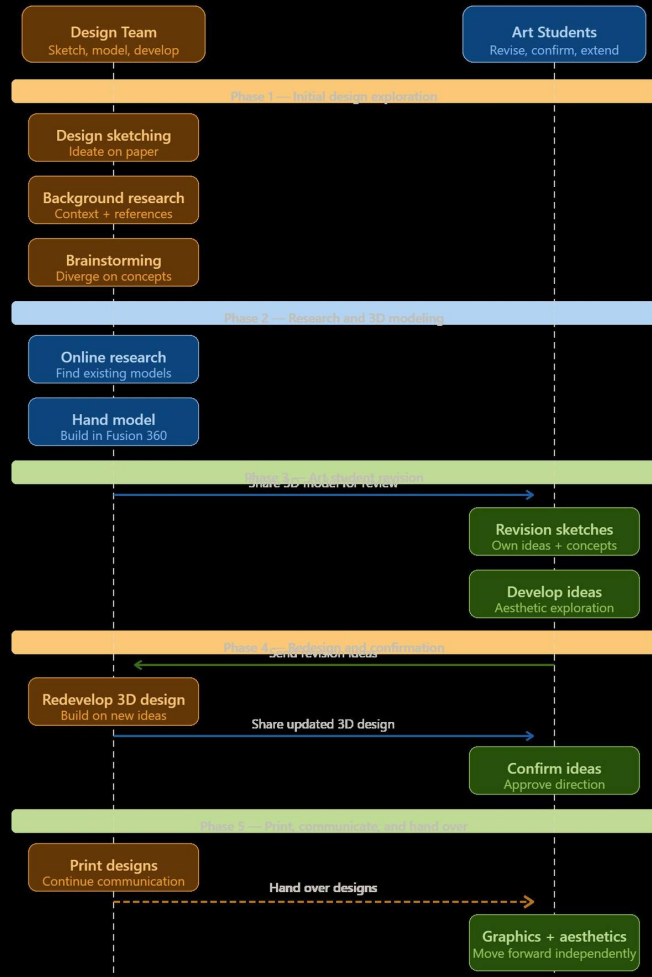
We learned how to use the fillet feature and communicate effectively

Future Applications:

- Electrodes (detect brain wave)
- Amplifiers (to boost signals)
- Filters to remove noise
- Microcontrollers to process data
- Bluetooth modules (for the data to be transmitted onto the app)
- Rechargeable battery
- Internal storage to log data



Final Design



References:

1. **Star 3d (Customizable) OpenScad Library.** *Thingiverse*, uploaded by mbeyerle116, 8 Nov. 2015, <https://www.thingiverse.com/thing:1118817/files>. Accessed 28 Apr. 2026.
2. **Montages and Technical Components.** *LearningEEG*, www.learningeeg.com/montages-and-technical-components (learningeeg.com in Bing). Accessed 28 Apr. 2026.
3. **EEG for Diagnosing Narcolepsy: What to Expect.** *MyNarcolepsyTeam*, www.mynarcolepsyteam.com/resources/eeg-for-diagnosing-narcolepsy-what-to-expect (mynarcolepsyteam.com in Bing). Accessed 28 Apr. 2026.